

phitron optional course

Math Basic Programming

Q About Division

$$\begin{array}{r} \downarrow \\ \swarrow 5 \mid \overset{\downarrow}{17} \mid 3 \rightarrow \text{Quotient} \\ \text{divisor} \quad \underline{15} \quad \searrow \text{dividend} \\ \quad \quad \quad 2 \rightarrow \text{Remainder} \end{array}$$

→ divisibility check

- check the remainder is zero

$$12 \% 3 = 0 \rightarrow 12 \text{ divisible by } 3$$

- What do we get if we just do the division.

$$12 / 3 = 4 \rightarrow \text{integer} \rightarrow \text{divisible}$$

$$12 / 5 = 2.4 \rightarrow \text{not integer} \rightarrow \text{"not"}$$

Q Prime and Composite Number.

→ Prime: Only two factors, 1 & itself.

→ Composite: There exists at least one factor other than 1 and itself.

Q Even and odd.

→ divisible by 2 → ~~Even~~

→ In binary last digit 0 is even
1 is odd.

Q Floor, Ceiling, Round.

Floor: Nearest integer below.

Ceiling: Nearest integer above

Round: Nearest integer.

$$12.65 \Rightarrow \text{Floor} = 12$$

$$\text{Ceiling} = 13$$

$$\text{Round} = 13$$

$$9.21 \Rightarrow \text{Floor} = 9$$

$$\text{Ceiling} = 10$$

$$\text{Round} = 9$$

Q Divisor Counting

$\Rightarrow$ NAIVE

take all integers $(1 \text{ to } n)$ and test remainder is zero.

more
better $\Rightarrow$

We only need to check
 $2 \text{ to } n/2$

even
better

We do not need to test any
number greater than $\sqrt{n}$.

Q Prime number or not

Take number from 2 to $\sqrt{n}$

- If any of them divide n , not prime

$\Rightarrow$ Sieve of Eratosthenes

Step 1: Take a number (2) and find its multiple in the range.

Step 2: those multiple must be composite cross them out.

Step 3: process to the next number

Step 4: If it's prime number go to step 1 otherwise 3.

1	2	3	4	5	6	7	8	9X	10
11	12	13	14	15X	16	17	18	19	20
2X	22	23	24	25	26	27X	28	29	30

prime number:- 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

* $\sqrt{n}$ ମଧ୍ୟରୁ ଯେଉଁଠାରେ multiple check କରାଯାଏ ସେହି ସଂଖ୍ୟାଗୁଡ଼ିକ କାଟି ଦିଆଯାଏ. ଯଦି $\sqrt{n}$ ଯାଏଁ count କରାଯାଏ,

Q Exact Extract Digits from a num.

• Example : 237

$$\rightarrow 237 \% 10 = 7$$

$$\rightarrow 237 / 10 = 23$$

$\rightarrow$ Repeat the same process with 23.

$$i = 0 \rightarrow n \% 10 = 7$$

$$i = 1 \rightarrow (n / 10) \% 10 = 3$$

$$i = 2 \rightarrow (n / 100) \% 10 = 2$$

General formula :-

$$\text{floor} \left[\frac{n}{10^i} \right] \% 10$$

Q GCD $\rightarrow$ greatest common divisor.

Example : 6, 15

- Divisor of 15 : 1, 3, 5, 15

- Divisor of 6 : 1, 2, 3, 6

$\rightarrow$ common divisor : 1, 3

- GCD = 3 (greatest divisor)

$\Rightarrow$ co-prime : if $\text{gcd}(a, b) = 1$, a and b are co-prime.

GCD $\Rightarrow$ multiplication of common factor.

Q GCD - Use Euclidean Algorithm.

$$\checkmark \text{gcd}(a, b) = \text{gcd}(b, a \% b)$$

$$\checkmark \text{gcd}(p, 0) = \text{gcd}(0, p) = p$$

$$\text{Example: } \text{gcd}(15, 6) = \text{gcd}(6, 3)$$

$$\text{gcd}(6, 3) = \text{gcd}(3, 0) = 3$$

$$\text{Example: } \text{gcd}(15, 8) = \text{gcd}(8, 7) =$$

$$\text{gcd}(7, 1) = \text{gcd}(1, 0) = 1 \rightarrow \text{co-prime.}$$

Q LCM - Least common multiple

$\checkmark$ take multiple of the large number and see if the smaller one divided it.

$$\Rightarrow \text{lcm}(15, 6): 15 \text{ multiple} = 15, 30, 45, 60, \dots$$

check. $15 \times 2 = 30$ is divided by 6?

if yes, 30 is LCM.

Another way $\Rightarrow$

$$\checkmark \text{gcd}(a, b) \times \text{lcm}(a, b) = a \times b$$

$$\text{lcm}(15, 6) = \frac{15 \times 6}{\text{gcd}(15, 6)} = \frac{90}{3} = 30$$

Q Factorial.

✓ non-negative integers.

$$\checkmark n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 3 \times 2$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

✓ Exception $0! = 1$

? How many type rearrange batting order of 11 players.

$$11! = 39916800.$$

Q Matrix / Matrices

$$A_{ij} = A_{\text{row}} = \begin{bmatrix} -2 & 5 & 6 \\ -5 & 2 & 7 \end{bmatrix}_{2 \times 3}$$

↓ ↓
row column

Matrix sum $\Rightarrow$ Two matrices can be added if they have same number of row and column.

Matrix multiplication:- two matrices A and B can be multiplied if the number of ~~col~~ columns of A and the number of rows of B is equal.

Q Power and base.

Power: $b^x \Rightarrow b$ multiple x times.

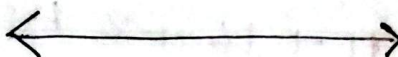
b - base

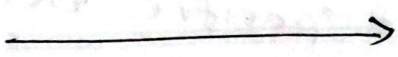
x - exponent

Root: $\sqrt[n]{a}$ (n th root of a)

$$\sqrt[n]{a} = x$$

$$\Rightarrow a = x^n$$

Q Line:  [Extended both side]

Ray:  [Extended one side]

Line segment:  [Finite length]

Q Right Angle ($\theta = 90^\circ$)

Acute Angle ($\theta < 90^\circ$)

Obtuse Angle ($\theta > 90^\circ$)

✓ Isosceles

✓ Equilateral

✓ Right-angle

✓ Scalene

└ Type of triangle

Q Distance between two point

$$d_{AB} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Q Series.

→ Sequence: List of number with non order or pattern.

→ 1, 3, 5, 7, 9

→ 1 1 2 3 5 8 13, ... [Fibonacci series]
sequence.

→ Series: Sum of element in a sequence.

$$S_1 = 1 + 3 + 5 + 7.$$

$$S_2 = 1 + 1 + 2 + 3 + 5 + 8 + \dots$$

→ Arithmetic series

$$1 + 6 + 11 + 16 + 21 + \dots$$

[Sum a specific number (5) in every step]

$$2 + 6 + 18 + 54 + 162 + \dots$$

→ Geometric Series

$$2 + 6 + 18 + 54 + \dots$$

[multiply by a specific number (3)
in every step]

Q Sum of natural number.

$$S = \frac{n(n+1)}{2}$$

→ Arithmetic series

$$n^{\text{th}} \text{ term} = a + (n-1) \cdot d$$

$$\text{sum of first } n \text{ terms} = \frac{n}{2} \{ 2a + (n-1)d \}$$

Q Unitary method

- Find value for single unit
- then multiply for necessary unit.

P. Tom finishes his homework within 15 hours while Jerry takes 10 hours. How many ~~100~~ hours will the same homework take to be done if they work together.

Idea: How many performance per hour.

Tom's 15 hour work $\equiv$ 1 homework

- Tom's 1 hour work $\equiv$ $\frac{1}{15}$ homework

Jerry 10 hour work $\equiv$ 1 homework

- Jerry's 1 hour work $\equiv$ $\frac{1}{10}$ homework

$\Rightarrow$ Tom and Jerry 1 hour combined
 $\equiv \left(\frac{1}{15} + \frac{1}{10} \right) = \frac{1}{6}$ homework

Answer : 6 hour.

Q Percentage

- Per hundred quantity.

P. A clothing store is selling one of the most popular products at 870 tk after 40% discount. What was the original price?

Let original price x .

the discounted price = $x \times 60\% = 870$

$$\Rightarrow x = \frac{870}{0.6} = 1450 \text{ tk.}$$

P. The price of oil increased by 25% and then decreased by 15%. What is the net percentage of increase or decrease in oil price?

Let's assume the initial price was 100.

increase 25% : updated price = 125

Decrease by 15% :

$$\rightarrow \text{updated price} = 85\% \times 125$$

$$= 106.25$$

$$\rightarrow \text{net price} = (106.25 - 100) = 6.25$$

Increased by 6.25%

☐ Capital and Interest

- Simple interest

$$I = Pnr$$

$$\text{Interest} = \text{Capital} \times \text{time unit} \times \text{Interest rate}$$

⇒ Your friend deposits 7000 tk in Sonali bank for 3 years which earn him an interest of 8%. What is the amount he gets after 3 years?

$$\begin{aligned} I &= Pnr \\ &= 7000 \times 3 \times \frac{8}{100} \\ &= \end{aligned}$$

⇒ Your deposit 5400 tk and got back an amount of 6000 tk after 2 years. Find the simple interest rate of the bank.

$$I = 6000 - 5400 = 600$$

$$n = 2$$

$$P = 5400$$

$$I = Pnr$$

$$r = \frac{I}{P \times n} = \frac{600}{5400 \times 2} \times 100\% = 5.56\%$$

Q) Mean & Median

→ Mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

→ Median:

✓ Sort them in ascending order

✓ If count is odd

→ Middle one is unique

✓ If count is even

→ there are two elements in the middle

→ take average of those two.

⇒ Why Median?

✓ Sometimes mean can be misleading.

Example:- 10 days history of your walking

3.5 km, 7, 5, 4, 4.5, 24, 5, 5.5, 5, 6.5

↓
outlier

→ mean is 7 km

without outlier it's 5.11 km

⇒ median

3.5, 4, 4.5, 5, 5, 5.5, 6.5, 7, 24

$$(5+5)/2 = 5$$

↓
not affect
in

Combinatorics

→ number of way to choose or arrange object

→ Arranging some object in all possible orders. $\Rightarrow$ factorial.

→ Choose some object out of a collection of object.

✓ combination problem

✓ how many way are there to choose r object from a collection of n object.

$${}^nC_r = ?$$

→ Choose and order some objects out of a collection of object

✓ permutation problem.

✓ How many way are there to arrange r object from a collection of n object.

$${}^nP_r = ?$$

$$* nPr = \frac{n!}{(n-r)!}$$

→ Batting order of 11 players cricket match from a squad of 15 players?

$$15P_{11} = \frac{15!}{(15-11)!} = \frac{15!}{4!}$$

$$* nCr$$

$$nC_r = \frac{nPr}{r!} = \frac{\frac{n!}{(n-r)!}}{r!} = \frac{n!}{r!(n-r)!}$$

$$\therefore nCr = \frac{n!}{r!(n-r)!}$$

$$15C_{11} = \frac{15!}{11!(15-11)!} = \frac{15!}{11! \cdot 4!}$$

▣ Probability

→ impossible event : 0

→ certain event : 1

$$P(\text{event}) = \frac{\text{Number of ways it can happen.}}{\text{Number of all possible outcomes}}$$

→ What is probability of getting a prime number if you throw a 6-faced die?

$$\text{target outcome} = 2, 3, 5 \quad \therefore P = \frac{3}{6} = \frac{1}{2} = 0.5$$

⇒ what is the probability of getting exactly two heads if you throw three coins?

$$\text{Number of possible outcomes} = 2 \times 2 \times 2 \\ = 8$$

Outcomes with exactly 2 heads out of 3 trials

- choose 2 positions out of 3 for H

$$- {}^3C_2 = \frac{3!}{2! 1!} = \frac{6}{2} = 3$$

$$\therefore P = 3/8$$